Activity 8.1: Coordinate Geometry

A

Name: _____ Partner's name: _____ Date: _____

Directions: Find the value asked for. Write points as ordered pairs. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

Find the distance between $(2, -1)$ and $(14, 4)$.

Answer: _____ Match? ☐ Yes ☐ No

2

Find the midpoint of the segment with endpoints $(-4, 2)$ and $(10, 8)$.

Answer: _____ Match? ☐ Yes ☐ No

3

$M(3, 5)$ is the midpoint of $\overline{AB}$ and $A = (1, 2)$. Find the coordinates of B.

Answer: _____ Match? ☐ Yes ☐ No

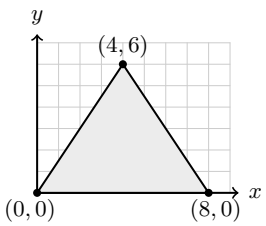
4

A line passes through $(0, 1)$ and $(2, 5)$. Find the slope of any line parallel to it.

Answer: _____ Match? ☐ Yes ☐ No

5

Find the area of the triangle.



Answer: _____ Match? ☐ Yes ☐ No

6

A rectangle has vertices $(0, 0)$, $(6, 0)$, $(6, 4)$, and $(0, 4)$. Find its perimeter.

Answer: _____ Match? ☐ Yes ☐ No

7

Find the perimeter of the triangle with vertices $(0, 0)$, $(6, 0)$, and $(3, 4)$.

Answer: _____ Match? ☐ Yes ☐ No

8

P is on $\overline{AB}$ with $AP : PB = 1 : 3$. $A = (0, 0)$ and $B = (8, 4)$. Find P.

Answer: _____ Match? ☐ Yes ☐ No

Activity 8.1: Coordinate Geometry

B

Name: _____ Partner's name: _____ Date: _____

Directions: Find the value asked for. Write points as ordered pairs. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

Find the distance between $(-3, 2)$ and $(9, 7)$.

Answer: _____ Match? ☐ Yes ☐ No

2

Find the midpoint of the segment with endpoints $(0, -1)$ and $(6, 11)$.

Answer: _____ Match? ☐ Yes ☐ No

3

The midpoint of $\overline{PQ}$ is $(4, 6)$ and $P = (3, 4)$. Find the coordinates of Q.

Answer: _____ Match? ☐ Yes ☐ No

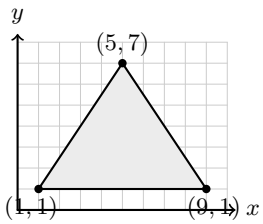
4

Line ℓ has slope $-\frac{1}{2}$. Find the slope of any line perpendicular to ℓ.

Answer: _____ Match? ☐ Yes ☐ No

5

Find the area of the triangle.



Answer: _____ Match? ☐ Yes ☐ No

6

A rectangle has vertices $(1, 1)$, $(9, 1)$, $(9, 3)$, and $(1, 3)$. Find its perimeter.

Answer: _____ Match? ☐ Yes ☐ No

7

Find the perimeter of the triangle with vertices $(0, 0)$, $(0, 6)$, and $(4, 3)$.

Answer: _____ Match? ☐ Yes ☐ No

8

P is on $\overline{AB}$ with $AP : PB = 1 : 3$. $A = (1, -1)$ and $B = (5, 7)$. Find P.

Answer: _____ Match? ☐ Yes ☐ No